ASMAN MAKERO MUSA

Computer Science Undergraduate & Applied AI Engineer

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PROFESSIONAL PROFILE

Analytical Computer Science undergraduate at the University of Nairobi with a rigorous mathematical foundation in linear algebra, multivariate calculus, and probability/statistics, combined with practical engineering experience in systems programming (C/C++) and modern machine learning pipelines. Experienced in designing Retrieval-Augmented Generation (RAG) engines, autonomous multi-agent decision copilots, and cloud integrations with schema-enforced tool calling. Demonstrated track record of building data-driven AI systems tailored to educational accessibility and operational challenges across Africa. Eager to advance machine learning competencies through the Carnegie Mellon University Africa AI Certificate.

EDUCATION & FOUNDATIONAL MATHEMATICS

University of Nairobi — Nairobi, Kenya *Expected Graduation:* ***Dec 2029***

Bachelor of Science in Computer Science *Current Status: Year 3*

- **Foundational Mathematics (CMU Prerequisites):** Linear Algebra (Vector spaces, matrix transformations, eigenvalues/eigenvectors), Calculus (Multivariate differentiation, optimization, partial derivatives), Probability & Statistics (Bayesian inference, probability distributions, hypothesis testing), Discrete Mathematics (Graph algorithms, combinatorics, Boolean logic).
- **Core Computer Science Foundations:** Data Structures & Algorithms (Algorithm complexity, dynamic programming, tree traversals), Low-Level Systems Programming (POSIX, pointers, memory hierarchies), Operating Systems (Processes, concurrency), Relational Database Management Systems (3NF normalization, SQL query optimization).

TECHNICAL SKILLS & DOMAIN EXPERTISE

AI & Machine Learning:	Retrieval-Augmented Generation (RAG), Multi-Agent Systems, LLM Benchmarking & Evaluation, Prompt Engineering, Function Calling / Tool Calling (JSON Schema), Vector Embeddings, Amazon Bedrock Foundation Models.
Programming Languages:	Python (FastAPI, PyPDF, Streamlit, Pandas, NumPy), C, C++ (Cisco Advanced Certified), SQL (PostgreSQL, Oracle, SQL Server), JavaScript / TypeScript (Next.js 15).
Databases & Vector Stores:	ChromaDB, Supabase (pgvector), PostgreSQL, Oracle Database, SQL Server, 3NF Relational Architecture, Data Pipeline Design.
Systems, Linux & Tools:	Red Hat Enterprise Linux (RH104), Bash Shell Scripting, Git/GitHub, REST APIs, JSON/YAML, CI/CD, Postman, SSMS, Zapier Automation.

APPLIED AI & SOFTWARE ENGINEERING PROJECTS

EduRAG — Academic AI Copilot & Multi-Unit Knowledge Retrieval Engine *Python, FastAPI, ChromaDB, Gemini Embeddings*

- Architected a unit-isolated Retrieval-Augmented Generation (RAG) study system to index complex university lecture slides, textbooks, and past papers across 7 heavy CS courses, eliminating generic LLM hallucinations via strict vector citation grounding.
- Implemented semantic chunking pipelines with header and slide-boundary awareness, tagging embeddings with structured metadata (Unit Code, Topic, Week, Document Name, Page Number) stored in ChromaDB vector collections.
- Engineered top-k cosine similarity retrieval with metadata pre-filtering and prompt synthesis to generate verified page-level answers, automated multi-choice practice exams with distractor justifications, and high-yield revision cheat sheets.

Agentic FPL Manager — Autonomous Multi-Agent Tactical Decision Engine *Python, Gemini 1.5 Pro, Game Theory, JSON Schema*

- Architected an autonomous multi-agent decision copilot featuring orchestrator and specialist analytical agents collaborating sequentially to model complex game-theory strategies under budget and formation constraints.
- Enforced strict JSON schema validation for Google Gemini 1.5 Pro tool calling, executing deterministic API queries against live football telemetry with zero syntax failure rates.
- Engineered mini-league rival intelligence algorithms computing Monte Carlo catch-up probability and modeling opponent chip usage patterns across a rolling 5-gameweek planning horizon.

APPLIED AI & SOFTWARE ENGINEERING PROJECTS (CONTINUED)

AzureVibe — Cloud Architecture & AZ-900 Readiness Platform *Next.js 15, FastAPI, Supabase, LLM Analogy Engine*

- Engineered an intelligent cloud exam preparation suite integrating active recall flashcards, 3D flip card mechanics, and Leitner spaced-repetition algorithms to accelerate cloud certification pass rates for African developers.
- Designed an AI Study Buddy leveraging customized system prompting to demystify abstract cloud architectures and distributed systems concepts through relatable African analogies (e.g., regional logistics corridors, agricultural cooperative pooling).
- Developed full-stack PostgreSQL/Supabase schema supporting real-time student quiz diagnostics, answer review history, and domain readiness scores.

Bundesliga Wrapped — Sports Analytics & Video Telemetry Pipeline *Python, Streamlit, YouTube Data API, REST*

- Developed an interactive sports analytics application consuming live match telemetry, standings, and highlight streams from Football-Data.org and YouTube Data API v3.
- Designed responsive data filtering, statistical aggregation charts, and deployed the application live on Streamlit Cloud.

Low-Level Systems Programming & Algorithm Optimization *C, C++, POSIX, Memory Management, Linux*

- Implemented core systems programming modules in C and C++, focusing on dynamic memory allocation, pointer arithmetic, custom data structure serialization, and algorithmic complexity optimization.
- Earned Cisco Certified Advanced C++ (CPA) credential, demonstrating mastery of Object-Oriented paradigms, Standard Template Library (STL), templates, exception safety, and low-level resource management.

PROFESSIONAL EXPERIENCE

Equity Bank Kenya — Nairobi, Kenya ***2024***

Intern — Operations, Credit Documentation & Customer Systems *Corporate Banking Division*

- Executed client financial documentation auditing and end-to-end loan credit validation workflows within a high-volume corporate banking environment.
- Collaborated with cross-functional branch teams utilizing enterprise CRM platforms and Microsoft Teams to streamline documentation pipelines and customer support operations.
- Applied rigorous analytical verification standards to guarantee 100% compliance with institutional regulatory requirements and internal record accuracy policies.

VERIFIED CERTIFICATIONS & ACCREDITATIONS

- **AWS & Udacity AI Practitioner Challenge:** Foundation models, Amazon Bedrock, LLM benchmarking & fine-tuning. *Udacity & AWS (2026)*
- **Cisco Certified Advanced C++ (CPA):** Advanced OOP, templates, memory management, STL algorithms. *Cisco & OpenEDG (2026)*
- **AI Safari — Generative AI & Prompt Engineering:** Multi-agent workflows, prompt design, AI engineering. *PLP & Emurgo Africa (2026)*
- **Red Hat Linux System Administration (RH104):** Linux CLI, POSIX process management, permissions, shell scripts. *Red Hat Academy (2025)*
- **Google Foundations of Cybersecurity:** Network defense, security architectures, threat mitigation. *Google & Coursera (2025)*
- **Cisco C++ Essentials 1 & 2:** Fundamental C/C++ control flow, memory handling, functional decomposition. *Cisco & OpenEDG (2025)*